

Assignment: 3

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Name: Kaushal Kumar Raj

Branch: B.Tech CSE AI & ML

PRN: 240205241023

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Subject: DBMS

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① Define transactions in DBMS.

→ A transaction is a sequence of one or more SQL operations treated as a single logical unit of work. It ensures data integrity - either all operations are executed successfully or none at all.

② List the ACID properties.

A - Atomicity: All or nothing execution

C - consistency: maintains database validity before and after execution.

I - Isolation: Transactions don't interfere with each other.

D - Durability: committed results persist even after a

(1) Explain the lost update problems.

③ What are the states of a transaction?

→ Active: Transaction is being executed.

Partially committed: All operations done, waiting to commit.

Committed: Successfully completed.

Failed: Error occurred before commit.

Aborted: Rolled back and all changes undone.

④ Explain serializability in transaction management.

→ Serializability ensures that concurrent transactions give the same result as if executed one after another. It's the main criterion for correctness in concurrent execution.

⑤ Differentiate between serial and serializable schedules.

→ serial: Transactions run one after the other without overlap
 serializable: Transactions interleave but yield the same result as a serial one
 serializable schedules improve concurrency while keeping consistency.

⑥ What is conflict equivalent schedule?

→ Two schedules are conflict equivalent if they have the same transactions and all conflicting operations occur in the same order.
 used to test conflict serializability in DBMS.

⑦ Explain recoverability in schedules.

→ A schedule is recoverable if a transaction commits only after all transactions it depends on have committed. It ensures that committed data isn't based on uncommitted or failed transactions.

⑧ What is a cascading rollback?

→ Occurs when a failed transaction causes others that read its uncommitted data to roll back too. This chain reaction can be avoided using cascadeless schedules.

⑨ Describe the two-phase locking protocol

→ In 2PL, transactions acquire locks in the growing phase and release them in the shrinking phase. Once a lock is released, no new locks can be acquired - ensuring serializability.

⑩ What are the type of locks used in concurrency control?

→ Shared (S): Allows multiple reads but no writes.
Exclusive (X): Allows one transaction to read/write.
Intention locks: Used in multi-granularity locking to indicate upcoming operations.

⑪ Explain the lost update problems.

→ Happens when two transactions read the same data and update it concurrently. One update overwrites the other, causing data loss - prevented by proper locking.

⑫ Describe the timestamp ordering protocol.

→ Each transaction is given a unique timestamp when it starts. Operations execute in timestamp order to ensure older transactions go first, maintaining serializability.

⑬ What is a view serializable schedule?

→ A schedule is view serializable if it produces the same final output as some serial schedule. It bases on data dependency and read-write relationships instead of conflicts.

⑭ What is a deadlock in transaction management?

→ Occurs when two or more transactions wait indefinitely for each other's back locks. Detecting using a wait-for graph and resolved by algorithms one transaction.

⑮ Explain the wait-die deadlock prevention scheme.

→ Older transactions can wait for younger ones, but younger ones requesting older locks are aborted ("die"). prevents circular waiting and hence avoids deadlocks.

⑯ Describe the rigorous two-phase locking protocol.

→ Locks (both read and write) are held until the transactions commits. provides strict serializability and prevents cascading rollbacks.

①7 What is a precedence graph?

→ A directed graph showing dependencies between transactions. If the graph has no cycles, the schedule is conflict-serializable.

①8 Explain the dirty read problem.

→ Occurs when a transaction reads uncommitted data from another transaction. If the other transaction rolls back, the first one has inconsistent data.

①9 What is a non-repeatable read?

→ A transaction reads the same row twice and gets different values because another transaction modified it. Prevented using Repeatable Read isolation level.

②0 Describe multi-version concurrency control (MVCC).

→ Maintains multiple versions of a data item. Readers access snapshot data while writers update new versions - improving concurrency.

②1 What are the isolation levels in SQL?

Read uncommitted: Allows dirty reads.

Read committed: No dirty reads.

Repeatable Read: Prevents non-repeatable reads.

Serializable: Highest level, avoids all anomalies.

Q22) What is cascadeless schedule?

→ A schedule where transactions only read committed data. Prevents cascading rollbacks and makes recovery simpler.

Q23) Explain the wound-wait deadlock prevention scheme.

→ If an older transaction needs a lock, held by a younger one, it "wounds" (aborts) it. If a younger one needs a lock from an older, it waits — preventing deadlocks efficiently.

Q24) Describe optimistic concurrency control.

→ Assume conflicts are rare; transactions execute without locks. Before commit, they validate to check for conflicts. If found, rollback occurs.

Q25) What is the phantom read phenomenon?

→ When a transaction reads a set of rows twice and finds new rows added or deleted by another transaction. Prevented using Serializable Isolation or Predicate locks.

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